

Tolerance response to water pH in larvae of two marine fish species, gilthead seabream, *Sparus aurata* (L.) and Senegal sole, *Solea senegalensis* (Kaup), during development

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Abstract

Gilthead seabream, *Sparus aurata* (L.) and Senegal sole, *Solea senegalensis* (Kaup) are two fish of primary importance in Mediterranean aquaculture. In the present study, the larvae of these species were exposed to different rearing-water pH during 24 h to examine their tolerance. The 24-h pHL50 values were calculated with low and high pH values at 7, 20 and 32 days after hatching (DAH) in *S. senegalensis* larvae, and 12, 20 and 52 DAH in *S. aurata* larvae. Low 24-h pHL50 values ranged between 4.88 and 5.76, whereas high 24-h pHL50 values ranged between 8.94 and 9.57 in *S. senegalensis* larvae. *S. aurata* larvae showed values of low 24-h pHL50 that ranged between 4.82 and 5.55, whereas values of high 24-h pHL50 ranged between 8.66 and 9.26. Both species showed similar tolerance response at all the tested ages. The high 24-h pHL50 values found were close to pH values that eventually can be reached in the rearing tanks. The pH should be carefully controlled in rearing water during the first development stages of both species.

Keywords: fish larvae, *Sparus aurata* (L.), *Solea senegalensis* (Kaup), tolerance response, pH, rearing water

Introduction

Most fish are very sensitive in their larval stages to any change in environmental conditions

(Westernhagen 1988). Changes in water characteristics can increase the mortality rate during the first developmental stages, therefore water quality is of special interest in aquaculture. The pH has been considered as an environmental factor that strongly affects the survival during the first developmental stages (Kamler 1992). Sea water has a high buffer capacity, and the pH is maintained in a narrow range (7.5–8.4). Rearing methods can induce changes in water pH beyond the expected normal values of natural sea water. For example, the oxidation process of the fish excretory products in recirculation systems induces a pH decrease (Brownell 1980). In contrast, microalgae routinely added to rearing tanks induces an increase in pH due to their CO₂ consumption.

The gilthead seabream, *Sparus aurata* (L.) and the Senegal sole, *Solea senegalensis* (Kaup) are two fish species of great importance for Mediterranean aquaculture. Larvae of these species are produced in the hatchery using live prey as food during the first weeks (Pascual & Yúfera 1987; Yúfera, Polo & Pascual 1993; Yúfera, Parra, Santiago & Carrascosa 1999). However, research on new rearing procedures using inert diets for this stage requires a better knowledge of larval tolerance to changes in water quality. Formulated diets are more susceptible to be attacked by bacteria and to generate pollutants in rearing water (Handy & Poxton 1993). Therefore, further knowledge of the tolerance limits of fish larvae to different abiotic factors is a required step when examining the potential for growth under different rearing conditions. The purpose of this paper is to define the 24-h mortality response in

larvae of gilthead seabream and Senegal sole when exposed to different pH values during development.

Materials and methods

The *S. aurata* and *S. senegalensis* eggs were obtained by natural spawning from a captive broodstock kept at a constant temperature of 19.5 °C. Different egg batches were used during these trials. The eggs were incubated at 19.5 °C and the newly hatched larvae were transferred to 300-L tanks with continuous illumination, water of 33‰ salinity and constant slight aeration. Initial larvae density ranged from 50 to 70 larvae per litre. The rotifers *Brachionus rotundiformis*, *Brachionus plicatilis* and *Artemia nauplii* were used as live food.

Sterilized seawater with 33‰ salinity and pH 8.2 was used for the tests. Analysis of the seawater showed low levels of ammonia-N and nitrite-N (0.32 p.p.m and 0.20 p.p.m, respectively). Modified sea water pH values for testing were obtained using Tris and phosphate-citrate buffer. Tolerance experiments were conducted with 12, 23 and 52 days after hatching (DAH) *S. aurata* larvae, and with 7, 20 and 32 DAH *S. senegalensis* larvae. The larval dry weights at the different ages are shown in Table 1. In the present study, larva response was analysed in a 24-h-test because in preliminary experiments we found that mortality increased rapidly when larva were not fed for more than 1 day.

All experiments were conducted with simultaneous controls, consisting of beakers with larvae taken in the same collection and using the above-mentioned sterile seawater. The pH was measured at the beginning and at the end of the experimental period, and the average pH value obtained was used for calculations. At least three replicates were conducted with each initial pH value. The lethal

pH values affecting 50% of the population in 24 h (pHL50) were determined from the linear regressions between the corrected mortality probits and log-transformed values of pH. Values for 95% confidence limits were also calculated in all cases (Abel & Axiak 1991).

Results

Larval mortality versus pH in *S. aurata* at 12, 23 and 52 DAH are shown in Fig. 1a–c. All of them showed a double-sigmoid response curve. Total mortality

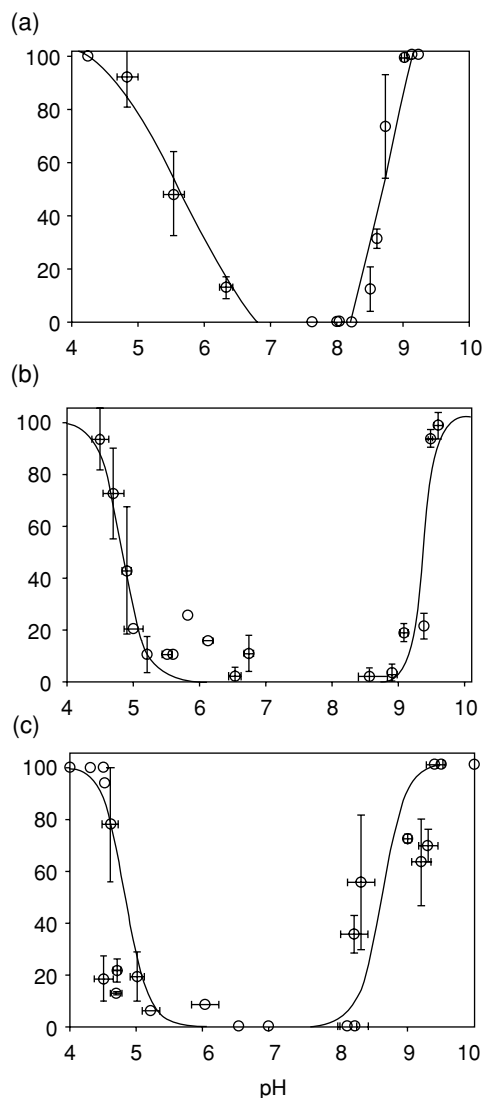


Figure 1 Corrected mortality in 12 (a), 23 (b) and 52 (c) DAH larvae of gilthead seabream, *Sparus aurata* (L.) after a 24-h test with different water pH.

Table 1 Summary of dry weights (mean and SD) of gilthead seabream, *Sparus aurata* (L.) and Senegal sole, *Solea senegalensis* (Kaup) larvae at different age used in the pH test

<i>Sparus aurata</i>		<i>Solea senegalensis</i>	
Age (DAH)	Dry Weight (µg)	Age (DAH)	Dry Weight (µg)
12	75 ± 14	12	113 ± 25
23	167 ± 40	22	1045 ± 250
52	1510 ± 320	32	1996 ± 399

occurred at pH values lower than 4.5 and higher than 9.5 at any tested age, excepting 12 DAH larvae, which reached total mortality at pH values over 9. The larval mortality probit exhibited a negative correlation with the log-transformed pH values at the low pH range, although this correlation was positive at high pH values Fig. 2.

The tolerance response to pH in *S. senegalensis* larvae at 7, 20 and 32 DAH is shown in Fig. 3a–c. As observed in *Sparus* larvae, mortality showed a double-sigmoid response curve with increasing water pH values. Total mortality occurred when pH was lower than 4.5 and higher than 9.5 at all tested ages. The correlation of probits versus log-transformed pH also was negative at the low pH range and positive at high pH values (Fig. 4). In both species, higher mortalities occurred in a narrower range at high pH values than at low pH values (Figs 2 and 4).

Table 2 shows the fitted parameters of the linear regressions of probits versus log-transformed pH. In *S. aurata*, no differences were found between the slopes obtained with low pH. At high pH, however, the slope obtained with 52 DAH larvae was significantly lower than those obtained with 12 and 23 DAH ($P < 0.05$). In *S. senegalensis*, no differences were found between the slopes obtained at different ages, in both low and high pH experiments. Values of 24-h low and high pH_{L50} in relation to larval dry

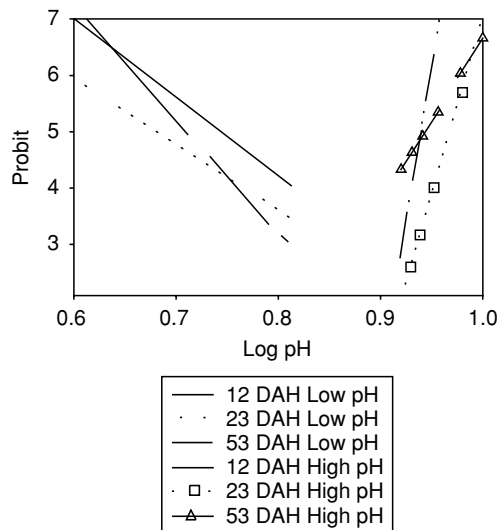


Figure 2 Regression lines obtained by probits transformation of mortality versus log-transformation pH in 12, 23, 52 DAH larvae of *Sparus aurata* after a 24-h test.

weight in *S. aurata* and *S. senegalensis* were similar in both species (Fig. 5).

Discussion

Both species showed the expected response shown by organisms exposed to an environmental variable that restricts survival within a limited range. These responses were in the form of a double-sigmoid curve when the mortality was plotted against the pH. The response obtained at high pH values was quicker than that at low pH (see Table 2 and Figs 4 and 5).

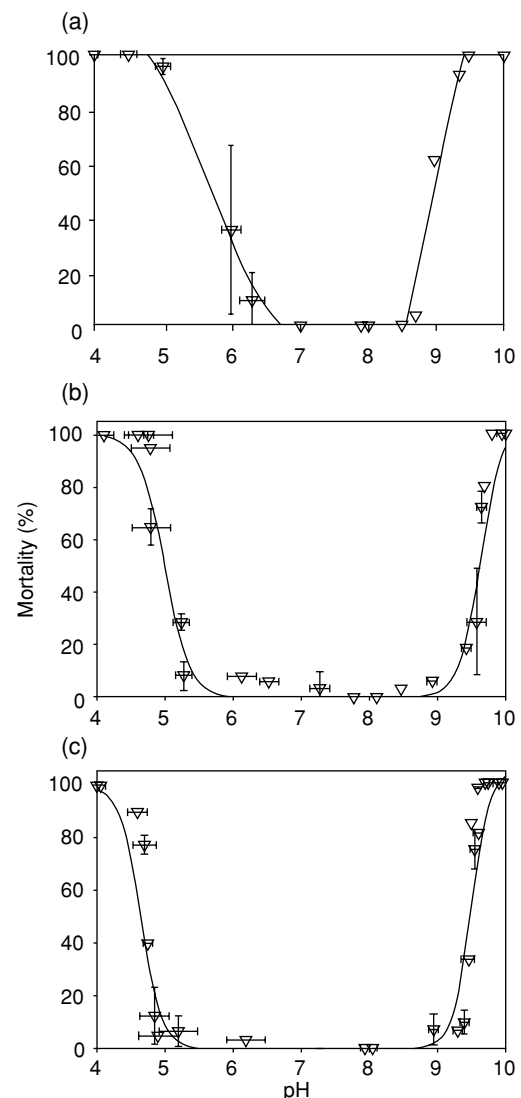


Figure 3 Corrected mortality in 7 (a), 20 (b) and 32 (c) DAH larvae of Senegal sole, *Solea senegalensis* (Kaup) after a 24-h test with different water pH.

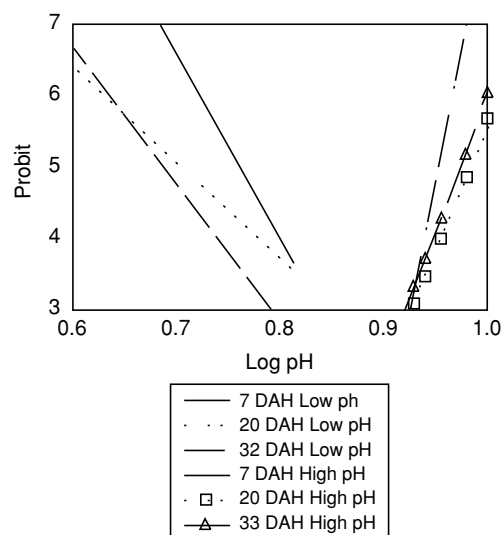


Figure 4 Regression lines obtained by probits transformation of mortality versus log-transformation pH in 7, 20, 32 DAH larvae of *Solea senegalensis* after a 24-h test.

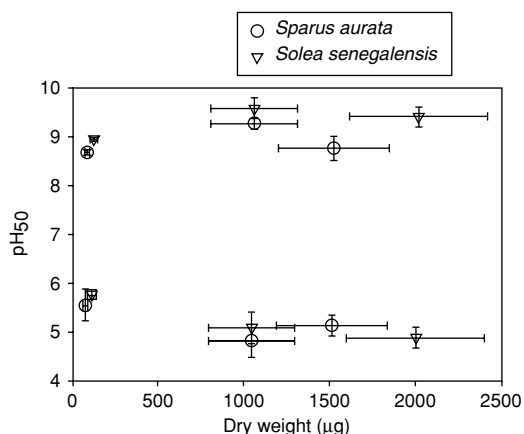


Figure 5 Values of pH₅₀ and 95% confidence limits versus larval dry weight (mean and SD) obtained in *Sparus aurata* and *Solea senegalensis*.

Table 2 Summary of tolerance regression models obtained in *S. aurata* and *S. senegalensis* larvae exposed to different pH values during larval development. The 24-h pH 50 values and the 95% confidence limits are included

	Age	Regression	24-h pH 50 (95% CL)
<i>S. aurata</i>			
Low pH	12	$Y = 15.28 - 13.81 X$ $r^2 = 0.73; n = 6; P = 0.0269$	5.55 (5.24–5.87)
	20	$Y = 113.01 - 11.73 X$ $r^2 = 0.63; n = 13; P = 0.0007$	4.82 (4.49–5.16)
	52	$Y = 19.34 - 20.19 X$ $r^2 = 0.69; n = 12; P = 0.0007$	5.13 (4.93–5.34)
High pH	12	$Y = -107.52 + 120 X$ $r^2 = 0.87; n = 7; P = 0.1061$	8.66 (8.60–8.72)
	20	$Y = -56.99 + 64.12 X$ $r^2 = 0.61; n = 11; P = 0.0043$	9.26 (9.14–9.38)
	52	$Y = -21.14 + 28.81 X$ $r^2 = 0.75; n = 14; P = 0.0005$	8.75 (8.50–9.00)
<i>S. senegalensis</i>			
Low pH	7	$Y = 24.50 - 25.65 X$ $r^2 = 0.73; n = 5; P = 0.0666$	5.76 (5.65–5.86)
	20	$Y = 14.52 - 13.49 X$ $r^2 = 0.67; n = 9; P = 0.0077$	5.08 (4.78–5.40)
	32	$Y = 18.19 - 19.16 X$ $r^2 = 0.56; n = 7; P = 0.0277$	4.88 (4.67–5.89)
High pH	7	$Y = -72.85 + 81.81 X$ $r^2 = 0.76; n = 4; P = 0.1366$	8.94 (8.90–8.98)
	20	$Y = -31.33 + 37.03 X$ $r^2 = 0.70; n = 10; P = 0.0030$	9.57 (9.36–9.79)
	32	$Y = -32.78 + 38.83 X$ $r^2 = 0.50; n = 10; P = 0.0280$	9.40 (9.20–9.60)

The ranges of water pH tolerance found in this study were wider than the regular pH values in open sea. *S. aurata* and *S. senegalensis* showed a similar response to pH changes and also similar to those reported by Brownell (1980) for other marine fish larvae at first feeding. In contrast, larvae of *S. senegalensis* exhibit higher tolerance to certain toxics such ammonia and nitrite than *S. aurata* larvae (Parra & Yúfera 1999).

The effect of extreme low pH values has received an extensive coverage in the literature. Low pH induces changes in the calcium metabolism, alters the osmotic activity, retards the spawning and reduces the hatching rate of fish (Westernhagen 1988; Duis & Oberemm 2000; Kitamura & Ikuta 2000). Nevertheless, studies on the effects of extreme high pH values are scarce. Its effect has been associated with a decrease in enzymatic activity because such extreme pH values are outside of the optimum range of enzymatic activity. Kamler (1992) reported a decrease in the chorionase activity at extreme pH levels, resulting in a hatching delay. Moreover, the high pH values affect osmoregulation, inducing a decrease in the permeability to Na^+ and Cl^- in the gills and to H^+ and HCO_3^- in the integument (Albers 1970). All these ions are essential in osmoregulation and in intra- and extra-cellular pH control. To compensate for those alterations, fish larvae have to invest more energy, which is diverted from the energy allocated for development, hence possibly affecting growth and survival. Rosenthal & Alderdice (1976) have noted that specific environmental conditions can induce smaller larvae, as a result of the increase in the maintenance costs, at the expense of metabolism contributing to growth.

Values for high pH₅₀ observed in the present study are very close to those normally found in larval rearing systems. Therefore, small fluctuations in the water pH can produce development problems and mortalities. In addition, pH changes are associated with the increase in the effect of other toxicants, such ammonia or heavy metals (Meade 1985; Sayer, Reader & Dalziel 1993).

When fish are exposed to fluctuating toxicant levels, the resulting toxicity may be higher than expected from the continuous exposure to LC₅₀ values, and more difficult to predict (Brown, Jordan & Tiller 1969). In this sense, the effects of fluctuating pH will show much lower tolerance levels than observed here. Brownell (1980) reported that larvae could present higher tolerance response to high pH

values if the exposure occurs in a gradual manner. The reason being that the major structures responsible for osmoregulation (gill, kidney, digestive tract) are not present in early larval stages and only appear later in development (Tytler, Bell & Robinson 1993).

Regarding the results obtained in the present study, toxic levels of pH can accidentally occur in the larval rearing tanks. Taken into account the 95% confidence limits, water pH values over 8.6 and 8.9 are not recommended in the rearing of early stages of gilthead seabream and Senegal sole respectively.

Acknowledgments

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